



120 TOPS AI accelerator module

The M.2 AI Edge accelerator module launched by Axelera sets the benchmark for superior performance as the cornerstone of artificial intelligence (AI) acceleration at the edge. Driven by a single Metis AIPU, the module achieves an impressive processing power of up to 120 tera-operations

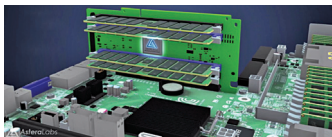


per second (TOPS) while ensuring minimal power consumption and seamless integration. With an affordable price tag, the module democratises AI processing, making it accessible to a wide range of applications. The module incorporates dedicated AI accelerators such as neural processing units or field-programmable gate arrays (FPGAs), significantly improving AI processing performance. The AI Edge accelerator reduces the need for continuous cloud connectivity by processing AI tasks locally, ensuring data privacy and security. The module has a small and standardised form factor, allowing for easy integration into various edge devices.

Axelera
<https://www.axelera.ai>

Memory connectivity platform for generative AI

Astera Labs has introduced the Leo CXL Memory Connectivity Platform to overcome memory bottlenecks and capacity limitations for artificial intelligence (AI) and machine learning (ML)



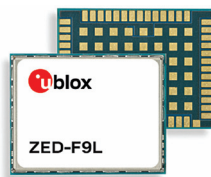
workloads. The memory connectivity platform is a solution that uses CXL 1.1 and 2.0 technologies to enable memory extension, pooling, and sharing. It stands as the industry's first purpose-built solution that offers end-to-end security protections, powerful fleet management tools, server-grade

customisable reliability, availability, and serviceability as well as seamless compatibility with top CPU, GPU, and memory suppliers. The platform addresses the demanding computational requirements of generative AI workloads with low latency such as ChatGPT. The connectivity platform supports high-speed data transfer rates, enabling efficient and fast communication between memory modules and other components. The platform offers scalability options, allowing for adding or expanding memory modules as needed.

Astera Labs
<https://www.asteralabs.com>

High-precision GNSS positioning modules

u-blox has launched two new global navigation satellite system (GNSS) modules, NEO-F9P and ZED-F9P-15B, based on their F9 high-precision GNSS platform. The modules achieve reliable positioning and attitude estimation, maintaining accuracy even



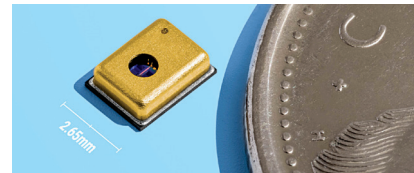
when GNSS services are unavailable, by integrating GNSS measurements, inertial measurement unit (IMU) data, wheel ticks, and vehicle dynamics. The modules ensure centimetre-level precision within seconds and come in an incredibly compact high-precision form factor, smaller than other such modules. The GNSS modules are designed to have a fast time to first fix, meaning they can quickly acquire satellite signals and calculate accurate position information. They are typically designed with low power consumption and are beneficial for battery-powered devices or applications.

u-blox
<https://www.u-blox.com/en>

Low-noise directional MEMS microphone

The SKR0410, a pulse density modulation (PDM) micro-electromechanical systems (MEMS) directional microphone launched by soundskrit, is the

industry's lowest noise PDM directional MEMS microphone. It is equipped with a dipole pickup pattern that offers superior signal isolation and successfully filters out acoustic background noise. The microphone reproduces sound with rich, high-fidelity due to its

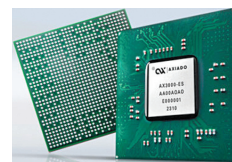


highly directed pickup pattern across the audible signal range. Even in noisy situations, its directed nature allows for the clear acquisition of the desired signal while minimising the need for intensive signal processing. The microphone is power-efficient due to its low current consumption of 685mA. Its small size of 2.65mm x 3.5mm x 1.1mm makes integrating into various gadgets and applications simple.

soundskrit
<https://soundskrit.ca>

AI-driven hardware security processors

The AX3000 and AX2000 trusted control/compute units (TCUs) launched by Axiado Corporation are the world's first fully integrated hardware security platform solutions driven by artificial



intelligence (AI). The processors' main function is to spot cyber security risks

and ransomware attacks on sophisticated servers and network switches in cloud data centres and 5G networks. The TCUs deliver a lower cost of ownership than any other alternative in the market. The TCUs provide a robust defence against side-channel attacks, such as differential power analysis, voltage glitching, and clock manipulation, which are utilised for extracting cryptographic keys. The company introduces a new class of cyber security processors equipped with forensic capabilities.

Axiado Corporation
<https://axiado.com>